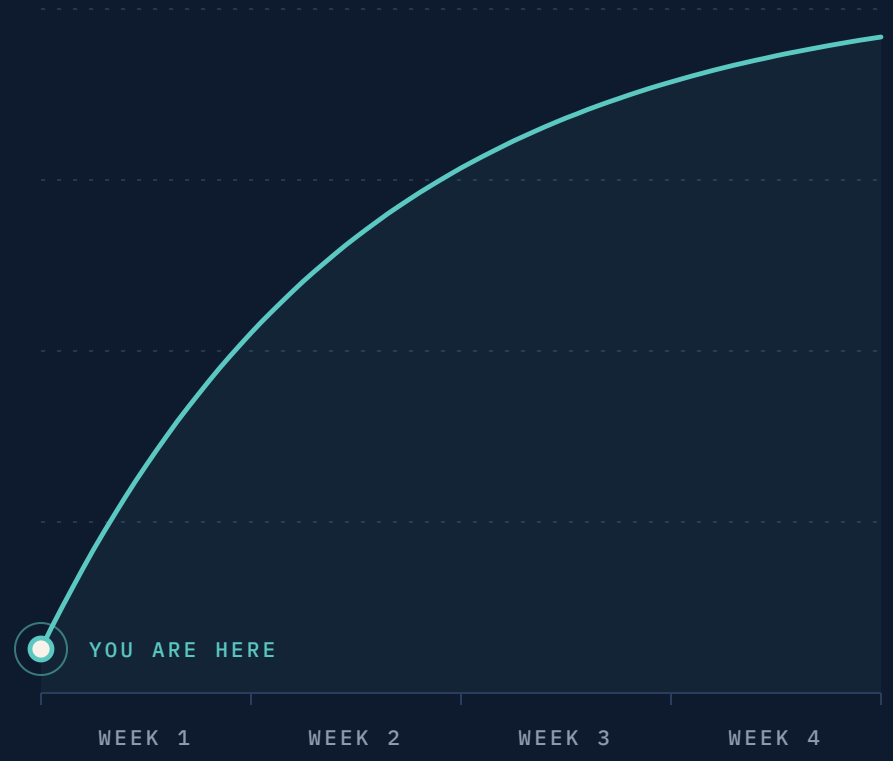
WEEK 1 · DAY 0 · OVERVIEW

Start Here

Python, Notebooks, and Libraries

From your first line of code to your own AI agents.
Every lesson is a fixed income problem.



Created by **Jeff Lenamon**

ABOUT 55 MIN WITH EXERCISES / 8 SECTIONS / 6 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The Morning Note

NOTE FROM THE DESK LEAD — 0 OF 2 ANSWERED

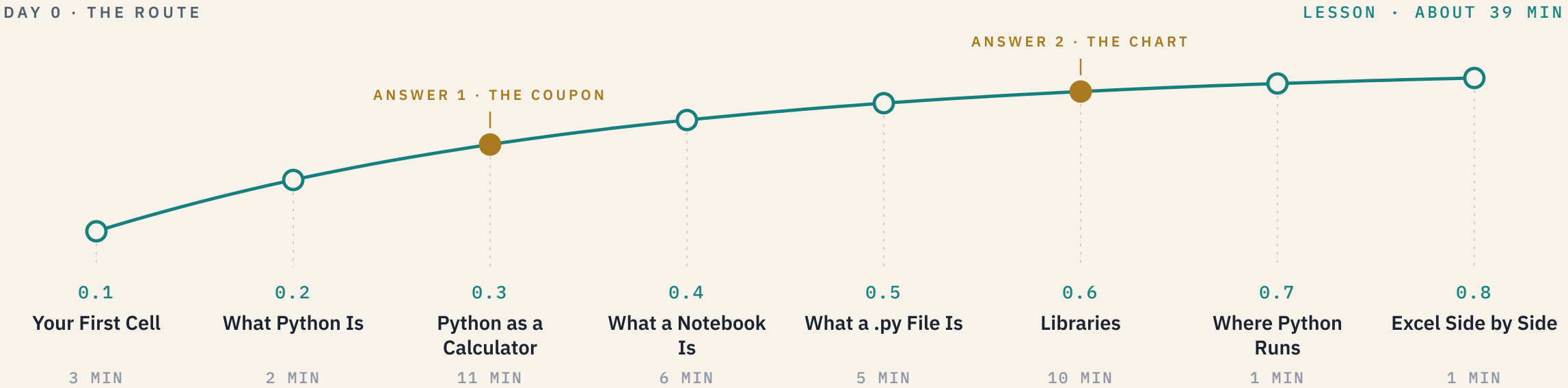
Before the 9:00 meeting, tell me what a 5.25 percent coupon pays each half year, and send me a picture of our three bonds.

- [ ] What a 5.25 percent coupon pays each half year
- [ ] A picture of our three bonds

You have never written a line of code. By the end of this notebook, Python will have answered both.

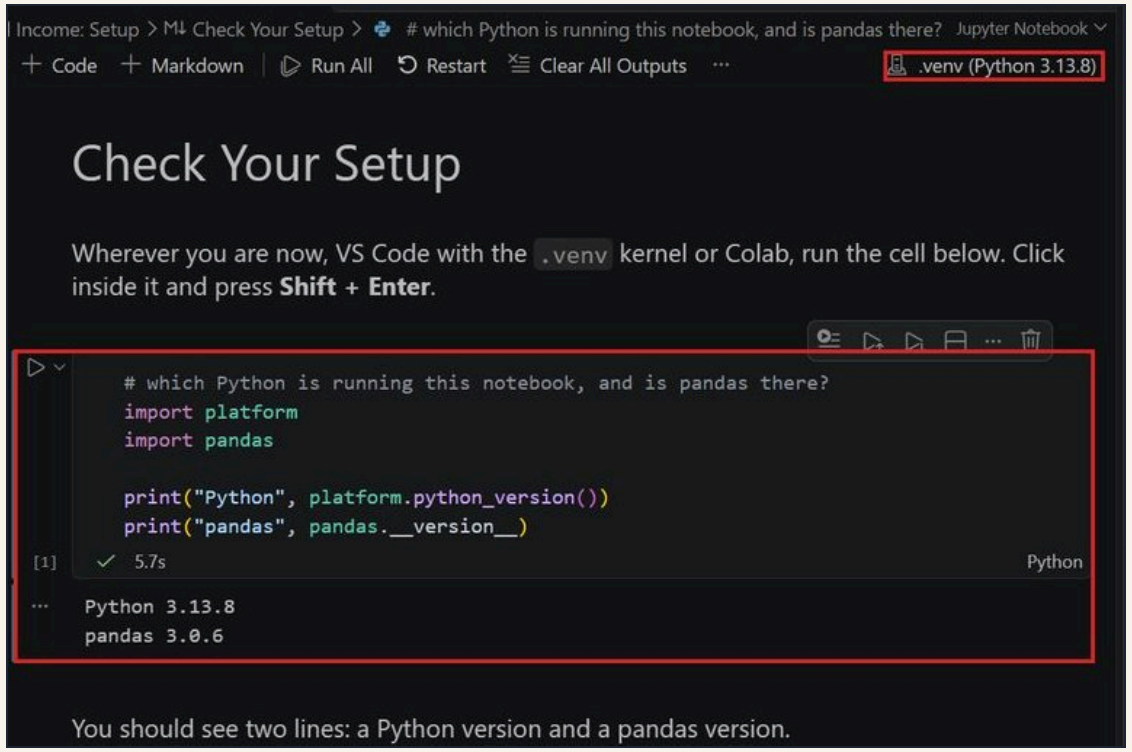
ABOUT 55 MIN WITH EXERCISES / 8 SECTIONS / 6 EXERCISES

Your Route Through Day 0



Time. About 39 minutes for the lesson, then about 18 for the six exercises.

0.1 Your First Cell



ONE CELL

```
# a line that starts with # is a comment
print("Good morning, bond desk")
```

```
Good morning, bond desk
```

Click in it and press **Shift + Enter**, or click its **play** button.



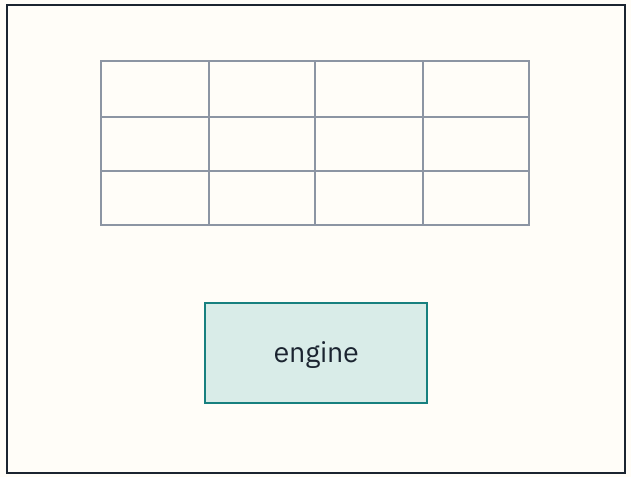
EVERY CELL, FROM THE TOP

Run All. Colab: **Runtime**, then **Run all**, or **Ctrl + F9**. VS Code: **Run All** at the top.

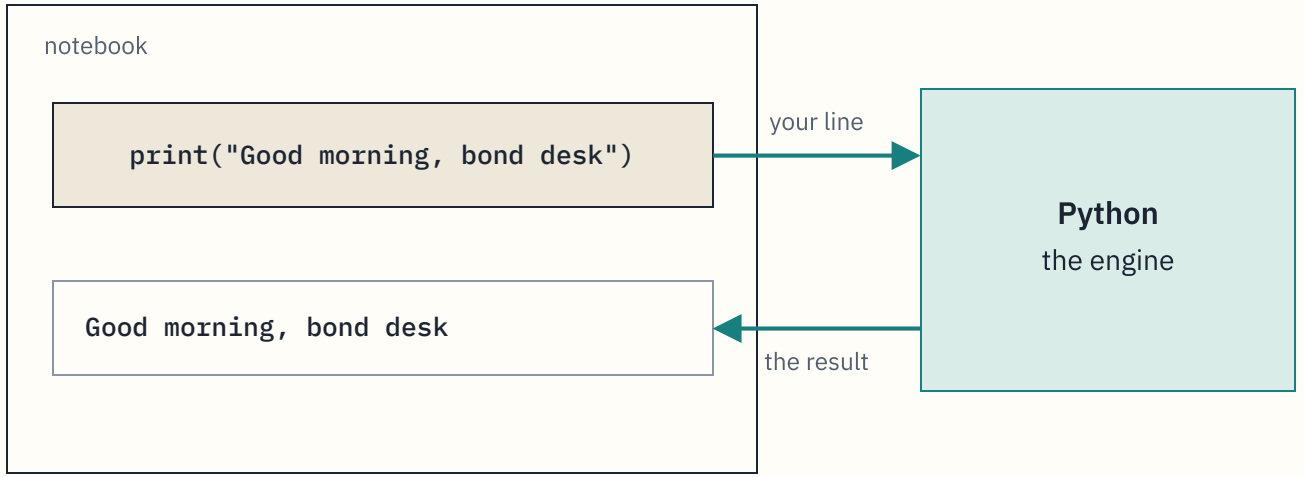
> **Something read that line and carried it out. That something is Python.**

0.2 What Python Is

Excel: grid and engine in one



Python: a notebook passes each line to the engine



EXCEL SIDE BY SIDE

Excel is the grid and the engine in one application. Python is only the engine.

WHY "PYTHON"?

Guido van Rossum started Python at CWI in the Netherlands over the 1989 Christmas holidays. The name has nothing to do with snakes: he was reading the scripts of *Monty Python's Flying Circus*, the BBC comedy, and wanted a name that was short, unique, and slightly mysterious.

Sources: the Python FAQ and "History of the software", read 25 October 2026.

0.3 Python as a Calculator

PYTHON	ANSWER	EXCEL
`2 + 2`	4	`=2+2`
`5 - 3`	2	`=5-3`
`5 * 5`	25	`=5*5`
`10 / 5`	2.0	`=10/5`
`10 // 5`	2	`=QUOTIENT(10,5)`
`7 // 2`	3	`=QUOTIENT(7,2)`
`7 % 2`	1	`=MOD(7,2)`
`5 ** 2`	25	`=5^2`

EXCEL SIDE BY SIDE

Each symbol is an **operator**, and a Python sum is an Excel formula without the `=`.

`/` always gives a decimal: `10 / 5` is `2.0`.

One Excel habit gives a wrong answer here, with no warning.

COMMON MISTAKE — A POWER TYPED THE EXCEL WAY

YOU TYPED

`5 ^ 2` *gives* 7

No error message: a wrong answer that looks like a right one.

TYPE THIS INSTEAD

`5 ** 2` *gives* 25

In Python, two stars make a power.

Three Habits, Then the Desk's First Sum

BRACKETS FIRST

```
2 + 3 * 4
```

```
14
```

```
(2 + 3) * 4
```

```
20
```

When in doubt, add brackets.

NO COMMAS IN NUMBERS

```
1,000,000
```

```
(1, 0, 0)
```

```
1_000_000
```

```
1000000
```

Python reads straight past underscores: they are only for your eyes.

HALF THE COUPON

```
5.25 / 2
```

```
2.625
```

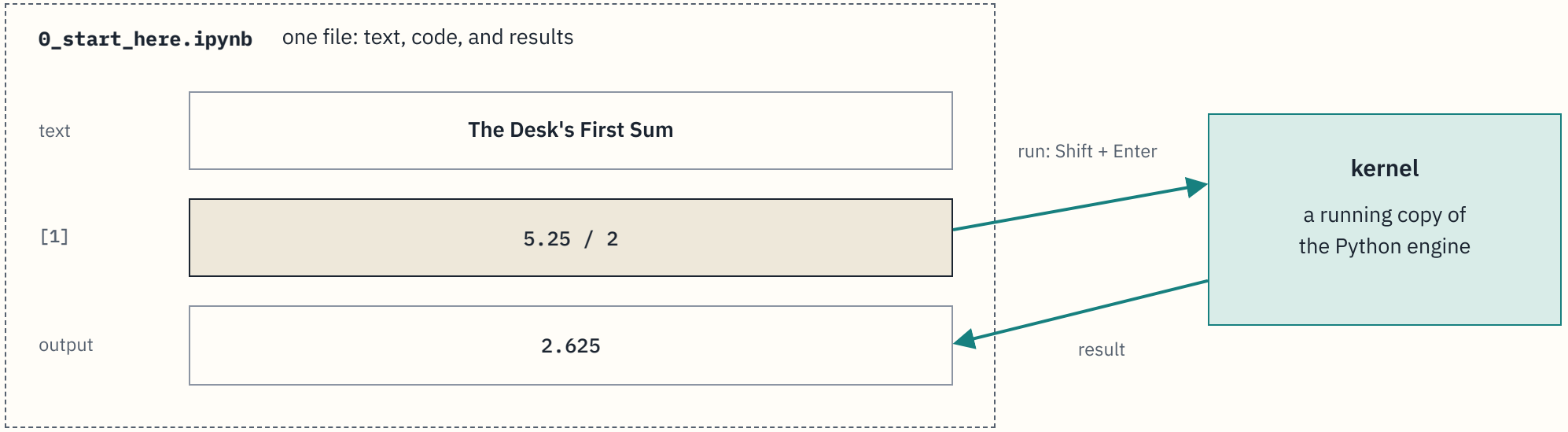
2.625 percent of face each half year: half the note answered.

THE MORNING NOTE

1 OF 2 ANSWERED

- [x] What a 5.25 percent coupon pays each half year `2.625 percent of face`
- [ ] A picture of our three bonds

0.4 What a Notebook Is



A CELL REMEMBERS

```
coupon = 5.25
coupon / 2
```

```
2.625
```

IN THE ORDER YOU RAN THEM

The kernel remembers in the order you **ran** the cells, not their order on the page.

If things get confusing, **Restart** (the kernel forgets everything), then **Run All**.

ADD A CELL

Rest the mouse between two cells, and two buttons appear.

Colab: **+ Code** or **+ Text**, also at the top of the page.

0.5 What a .py File Is

1. MAKE THE FILE

```
%%writefile half_coupon.py
# the annual coupon, in percent
coupon = 5.25
# print shows the label, then half the coupon
print("Half the coupon:", coupon / 2)
```

```
Writing half_coupon.py
```

2. LOOK AT THE FILE



3. RUN THE FILE

```
%run half_coupon.py
```

```
Half the coupon: 2.625
```

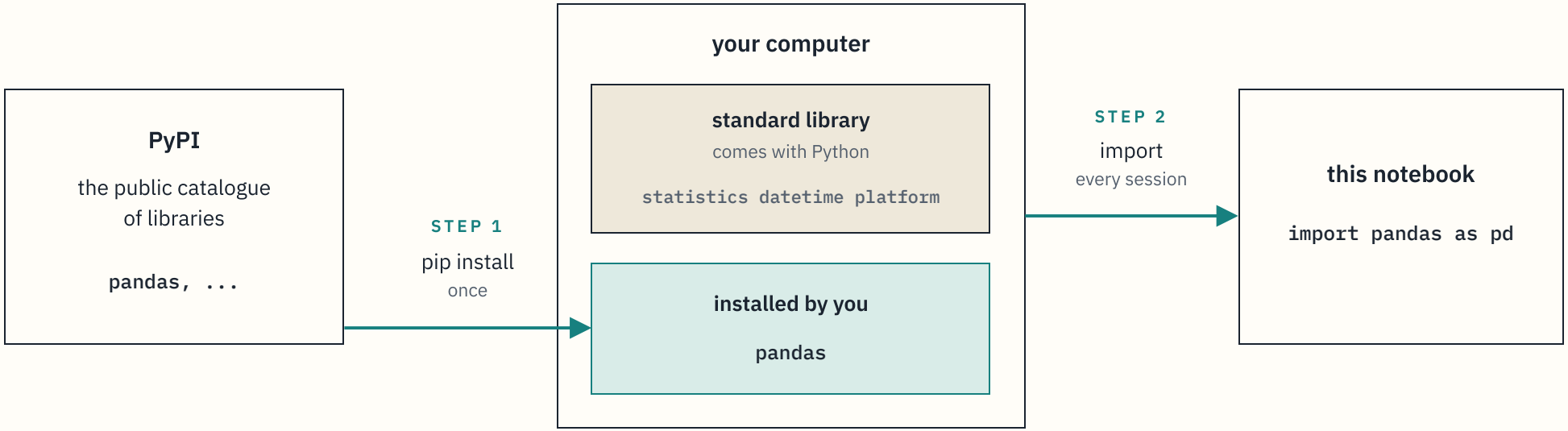
Outside a notebook, the same file runs from a **terminal**, a window for typing commands, with `python half_coupon.py`.

4. CHANGE THE FILE

1. In the cell that starts `%%writefile`, change `5.25` to `6.5`.
2. Run that cell. It answers `Overwriting half_coupon.py`: the file was already there, and now holds the new number.
3. Run the `%run half_coupon.py` cell again.

You should see `Half the coupon: 3.25`.

0.6 Libraries



BUILT IN

```
prices = [98.50, 101.25, 99.75]
print("Highest:", max(prices))
```

```
Highest: 101.25
```

STATISTICS

```
import statistics
statistics.mean(prices)
```

```
99.83333333333333
```

DATETIME

```
from datetime import date
days = date(2026, 7, 15) - date(2026, 1, 15)
print(days.days, "days")
```

```
181 days
```

Three Bonds in a Table, Then the Picture

THREE BONDS IN A TABLE

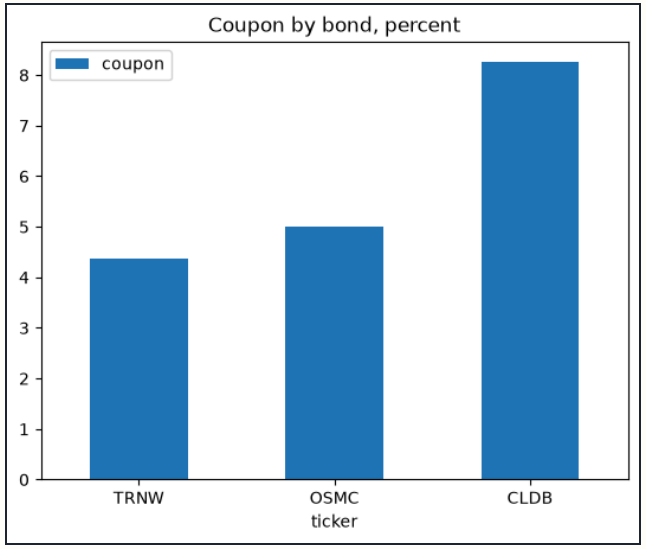
```
import pandas as pd
bonds = pd.DataFrame({
    "ticker": ["TRNW", "OSMC", "CLDB"],
    "coupon": [4.375, 5.00, 8.25],
    "price": [96.25, 100.00, 104.50],
})
```

A LIBRARY THAT IS NOT THERE

```
import bondtools
```

```
ModuleNotFoundError: No module named 'bondtools'
```

ONE MORE LINE: THE PICTURE




```
chart = bonds.plot.bar(x="ticker", y="coupon",
    rot=0, title="Coupon by bond, percent")
```

THE MORNING NOTE

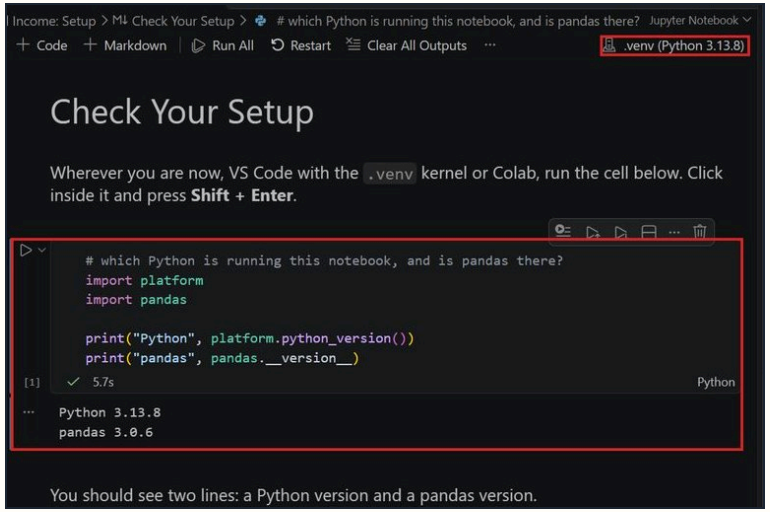
2 OF 2 ANSWERED

- [x] What a 5.25 percent coupon pays each half year — `2.625 percent of face`
- [x] A picture of our three bonds — `the coupon chart`

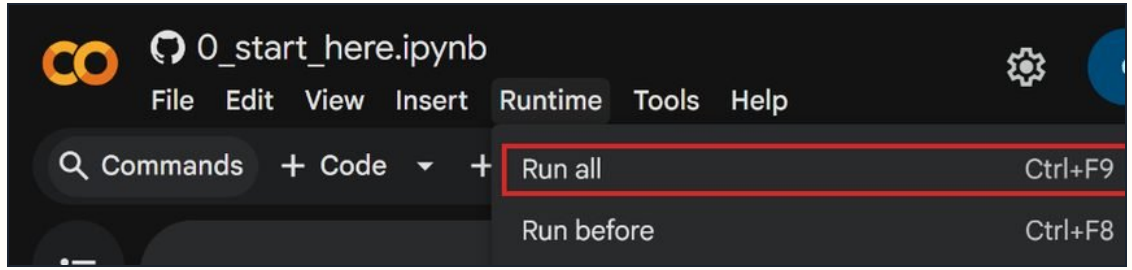
0.7 Where Python Runs

	VS CODE	COLAB
Runs on	your Windows computer	Google's computers, in a browser
Libraries live in	a **virtual environment**: a `.venv` folder with its own Python and libraries	Google's computers
Data	on a work computer, your firm's rules come first	public or made-up data only

VS CODE



COLAB



The Setup notebook walks through both, step by step.

0.8 Excel Side by Side

EXCEL	PYTHON
A workbook (`.xlsx`)	A notebook (`.ipynb`)
Excel, the application that calculates	Python, the engine
A formula, `=5.25/2`	A code cell, `5.25 / 2`
A text box on the sheet	A markdown cell
Pressing F9	Run All
`COUNT`, `SUM`, `MAX`, `MIN`, `AVERAGE`	`len`, `sum`, `max`, `min`, `statistics.mean`
An add-in, such as the Analysis ToolPak	A library
A VBA module	A `.py` file

One big difference: Excel, on its default automatic setting, recalculates by itself; a Python cell runs only when you run it.

You Can Now

- Run a cell, or the whole notebook with Run All.
- Say what Python is: a language, and the engine that runs it.
- Use Python as a calculator, with `**` for a power, never `^`.
- Tell markdown cells from code cells, and add both.
- Make a `.py` file, run it, and say how it differs from a notebook.
- Install and import a library, build a table, draw a chart from it, and read a missing-library error.

THE MORNING NOTE — 2 OF 2 ANSWERED

- [x] What a 5.25 percent coupon pays each half year — `2.625 percent of face`
- [x] A picture of our three bonds — `the coupon chart`

NEXT

WEEK 1 · DAY 1, PART 1

Python for Fixed Income: Introduction, Part 1

You meet the whole desk: four bonds, and your first real calculations on them.